Kshitij Hedau

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github.com/Herostomo

Resume

I am **Kshitij Hedau**, an undergraduate student at **PCCoE (Pimpri Chinchwad College of Engineering)** pursuing a degree in **Electronics and Telecommunication Engineering**, with a strong interest in digital design and embedded systems. Skilled in **Verilog, FPGA development, RTL design, and SystemVerilog for verification**, with hands-on experience in hardware-software co-design and embedded systems. Passionate about building efficient, scalable hardware solutions and bridging the gap between electronics and modern computing systems.

Education

Pimpri Chinchwad College of Engineering (PCCoE), Pune *2023 – Present*
B.E. in Electronics and Telecommunication Engineering CGPA: 8.7 / 10 (till 6th semester)

Experience

Technical Assistant Intern at Innovative Solutions Jan 2026 – March 2026

- Developed an Automated Vehicle Speed Alert Testing and Compliance System using ESP32 and PLC for beep testing, signal detection, and industrial automation.

Projects

AI Scribe : AI Scribe is an assistive system for dyslexic students that converts spoken answers into written text on paper using voice recognition and a 2D plotting mechanism, providing an independent alternative to a human scribe during CBSE exams.

Automated Vehicle Speed Alert Testing and Compliance System : Built a proof-of-concept embedded system to identify vehicle speed warnings by analyzing beep frequencies and timing patterns using an ESP32 and digital microphone. Designed dual-mode firmware and hardware switching for multiple automotive standards.

Custom ML Instruction on RISC-V - GitHub
Implemented a 5-stage pipelined RISC-V processor in Verilog and extended the ISA with a custom ML instruction (`is_softmax`) to enable ML computation at the hardware level. Modified decode logic, pipeline registers, and execution unit to integrate the custom datapath. Designed register file, instruction memory, and control flow, and validated functionality using ModelSim and GTKWave through waveform analysis.

Shor's Algorithm Implementation using IBM Qiskit
Designed and implemented a quantum computing solution for integer factorization using Shor's Algorithm on the Qiskit framework. Developed quantum circuits for modular exponentiation and quantum phase estimation, enabling efficient factorization of small composite numbers on simulators and IBM Quantum backends. Analyzed quantum circuit depth, gate complexity, and noise impact on real hardware, and applied optimization techniques to improve execution fidelity. Demonstrated the implications of quantum algorithms on classical cryptographic systems such as RSA.

Certifications

AI Programming with Python — Udacity View Certificate

FPGA Design with Verilog — (Udemy)

Technologies

Programming Languages: Python, C, C++, Embedded C
RTL Design: Verilog, SystemVerilog, Digital Design, RTL Design, FSM Design, Pipelining
Embedded Systems: Microcontrollers, Embedded C Programming, Peripheral Interfacing (UART, SPI, I2C)
FPGA Tools: Xilinx Vivado, ModelSim, Icarus Verilog, GTKWave,